Pets Trading System

- System Requirements
  - 1. Business Context / Purpose
  - 2. Core System Mechanics
    - 2.1 Traders
    - 2.2 Pets
    - 2.3 Market & Trades
    - 2.4 UI & Views
    - 2.5 System Behavior
  - 3. Optional / Bonus Features
  - 4. Ready-to-Use Pet Dictionary
    - Intrinsic Value Formula
  - 5. Example Flows
    - 5.1 Purchasing New Pets
    - 5.2 Secondary Market Trade
    - 5.3 Bid Withdrawal
    - 5.4 Valuation Update
    - 5.5 Bid Being Outbid
    - 5.6 Trader Delisting a Pet
    - 5.7 Reviewing Intrinsic Value
    - 5.8 Using Leaderboard
- Clarifying Questions & Answers
  - 1. Trader Behavior & Controls
  - 2. Market Mechanics & Trading Rules
  - 3. Pets & Valuation
  - 4. UI / Views
  - 5. System Behavior
- UX / Experience Considerations left to Participant Decisions
  - 1. Market View Display
  - 2. Trader Panel / Inventory
  - 3. Analysis / Drill-Down View
  - 4. Leaderboard
  - 5. Notifications
  - 6. Bid / Trade Interaction
  - 7. Listing / Delisting Pets
  - 8. Valuation Updates
  - 9. Multiple Active Panels
  - 10. Optional UX Enhancements

System Requirements

1. Business Context / Purpose

Participants will build a **"Trading Pets" system** to demonstrate **end-to-end AI-assisted system development**, including design, coding, deployment, and automated testing.

- **Goal:** Human-controlled Traders buy, sell, and manage virtual pets, demonstrating AI-assisted system development rather than trading strategies.
- **Focus Areas for AI Use:**
  - UI/UX design
  - Technical architecture / data modeling
  - Front-end & back-end coding
  - Cloud deployment (IaC)
  - Automated testing

---

2. Core System Mechanics

2.1 Traders

- **Number:** Exactly 3 Traders.
- **UI Panels:** Each Trader has a separate panel/window.
- **Private Information:** Inventory, available cash, locked cash, portfolio value, and notifications.
- **Cash:** Fixed initial amount sufficient to buy 5–8 new pets.
- **Portfolio Value:** `available cash + locked cash + market value of pets`.

2.2 Pets

- **Dictionary:** Provided read-only, 20 breeds (5 dogs, 5 cats, 5 birds, 5 fish).
- **Parameters per breed:**
  - Lifespan (years)
  - Desirability (numeric score)
  - Maintenance cost
  - Full health = 100%
  - Age = 0 (new pets)
- **Intrinsic Value Formula:** Same for all pets; uses breed-specific parameters.

$$\text{Intrinsic Value} = \text{Base Value} \times \left(\frac{\text{Health}}{100}\right) \times \left(\frac{\text{Desirability}}{10}\right) \times \left(1 - \frac{\text{Age}}{\text{Lifespan}}\right)$$

- **Supply:** Limited (default 3 per type), decreases as pets are purchased.
- **Unique Entities:** Each pet instantiated separately; lifecycle tracked individually.
- **Lifecycle Updates:** Age increases continuously; health and desirability ±5% per update. Updates every minute (configurable).

2.3 Market & Trades

- **New Pet Purchases:**
  - From supply at fixed retail price.
  - Multiple pets may be purchased if cash and supply allow.
  - Not considered a secondary-market trade.
- **Secondary-Market Trades:**
  - One pet per transaction.
  - Bid above or below asking price allowed.
  - Highest bid is active; cash locked.
  - Seller can accept/reject; trade executes immediately.
  - Bids > available (unlocked) cash are rejected.
  - Buyers cannot bid on their own pets.
  - Buyers only see status of their own bids (active, rejected, withdrawn, outbid).
- **Listings:**
  - Asking price > 0.
  - Pet listings can be withdrawn by seller; active bids are rejected and locked cash released.
  - Only one active listing per pet.
  - Multiple pets may be listed simultaneously.

2.4 UI & Views

- **Trader Panel:** Inventory, available cash, locked cash, total portfolio value.
- **Market View:**
  - Current listings
  - Asking price
  - Most recent trade price
  - New supply count
  - Default order = newest listings first (optional sorting/filtering)
- **Analysis / Drill-Down View:**
  - Full pet fundamentals (age, health, desirability, intrinsic value)

    - Expired status
- **Notifications:**
    - Bid received, accepted, rejected, withdrawn, outbid
    - Include pet, price, counterparty
    - Chronological order

2.5 System Behavior

- Sequential actions sufficient for demo; no concurrency enforcement required.
- Trades and valuation updates trigger immediate UI refresh if any metrics change.
- Expired pets remain in inventory; residual market value is market-driven.
- Cash for active bids locked; released upon bid withdrawal or rejection.
- Multiple active bids across different pets allowed.

---

3. Optional / Bonus Features

- Sorting/filtering in market view
- Bid timestamps for history
- Performance / scalability considerations
- Audit trails / ledger beyond notifications
- Enhanced UI interactions (confirmation prompts, visualization enhancements)

---

4. Ready-to-Use Pet Dictionary

Type	Breed	Lifespan	Desirability	Maintenance	Health	Age	Base/Retail Price
Dog	Labrador	12	8	5	100	0	100
Dog	Beagle	13	7	4	100	0	90
Dog	Poodle	14	9	6	100	0	110
Dog	Bulldog	10	6	7	100	0	80
Dog	Pit Bull	11	5	5	100	0	70
Cat	Siamese	15	9	4	100	0	90
Cat	Persian	14	8	6	100	0	85

Cat	Maine Coon	16	7	5	100	0	80
Cat	Bengal	12	6	5	100	0	75
Cat	Sphynx	13	5	7	100	0	70
Bird	Parakeet	8	7	3	100	0	25
Bird	Canary	10	6	2	100	0	20
Bird	Cockatiel	12	8	3	100	0	30
Bird	Macaw	50	9	8	100	0	120
Bird	Lovebird	15	5	3	100	0	15
Fish	Goldfish	10	5	2	100	0	5
Fish	Betta	5	6	1	100	0	6
Fish	Guppy	3	4	1	100	0	4
Fish	Angelfish	8	7	2	100	0	8
Fish	Clownfish	6	8	3	100	0	10

Intrinsic Value Formula

$$\text{Intrinsic Value} = \text{Base Value} \times \left(\frac{\text{Health}}{100}\right) \times \left(\frac{\text{Desirability}}{10}\right) \times \left(1 - \frac{\text{Age}}{\text{Lifespan}}\right)$$

Where:

- **Base Value** = Suggested retail price for the pet type (set in the dictionary)
- **Health** = Current health percentage (0–100%)
- **Desirability** = Breed-specific desirability (1–10 scale)
- **Age** = Current age of the pet in years
- **Lifespan** = Maximum lifespan of the breed in years

Notes:

- Age starts at 0 when purchased from supply.
- Health and desirability fluctuate ±5% each update (***once per minute***).

- Residual value at end-of-life is determined by the market (Traders can still bid on “expired” pets).

Here’s a **table of 20 random scenarios** that demonstrate the use of the formula to calculate the intrinsic value of a pet:

Type	Breed	Age (yrs)	Health (%)	Desirability	Base Price	Intrinsic Value
Cat	Sphynx	12.29	77.82	5	70	1.48
Dog	Poodle	0.77	94.24	9	110	88.16
Bird	Lovebird	2.17	55.95	5	15	3.59
Fish	Guppy	0.82	86.18	4	4	1.00
Fish	Goldfish	8.77	67.85	5	5	0.21
Dog	Labrador	3.12	92.11	8	100	61.20
Cat	Siamese	5.64	78.34	9	90	42.13
Dog	Beagle	6.45	84.22	7	90	27.08
Bird	Macaw	10.57	95.11	9	120	105.77
Fish	Betta	1.12	73.45	6	6	3.46
Dog	Pit Bull	8.22	67.83	5	70	5.12
Cat	Persian	4.91	88.33	8	85	38.22
Dog	Bulldog	2.78	95.44	6	80	44.01
Cat	Maine Coon	6.33	90.12	7	80	35.26
Bird	Cockatiel	3.44	77.22	8	30	15.56
Fish	Angelfish	5.12	69.81	7	8	1.72
Bird	Parakeet	1.55	82.45	7	25	14.34
Fish	Clownfish	2.78	88.11	8	10	5.02
Cat	Bengal	8.22	75.22	6	75	9.78
Dog	Poodle	10.11	85.33	9	110	25.27

This table demonstrates a **range of intrinsic values** for different pets at different ages and health levels, consistent with the formula.

---

5. Example Flows

5.1 Purchasing New Pets

1. Trader A buys 2 Labradors from new supply at retail price.
2. Cash decreases, pets added to inventory, lifecycle metrics start ticking.

5.2 Secondary Market Trade

1. Trader A lists a Poodle for sale at $Y.
2. Trader B places a bid of $Z > $Y.
3. Trader A accepts → trade executes immediately.
4. Notifications:
   - Trader A: “Bid accepted by Trader B for Poodle at $Z.”
   - Trader B: “Your bid accepted by Trader A for Poodle at $Z.”

5.3 Bid Withdrawal

1. Trader C bids $W on a Bengal.
2. Trader C withdraws bid → cash released.
3. Trader A notified: “Bid withdrawn by Trader C for Bengal at $W.”

5.4 Valuation Update

1. Every minutes, intrinsic value recalculated using ±5% variance.
2. All affected panels refresh automatically.

5.5 Bid Being Outbid

1. Trader A lists Bulldog at $50.
2. Trader B bids $55 → active.
3. Trader C bids $60 → replaces B’s bid, B’s cash released.
4. Notifications:
   - Trader A: “New highest bid $60 from Trader C.”
   - Trader B: “Your bid $55 outbid by Trader C.”
   - Trader C: “Your bid $60 is currently highest.”

5.6 Trader Delisting a Pet

1. Trader A has Poodle listed with active bid $40 from Trader B.

2. Trader A withdraws listing → bid rejected, cash released.
3. Notification: Trader B: “Bid $40 withdrawn by Trader A (listing removed).”
4. Pet returned to Trader A inventory.

5.7 Reviewing Intrinsic Value

1. Trader C views analysis for Cocker Spaniel listed by Trader B.
2. Analysis shows age, health, desirability, maintenance, intrinsic value.
3. Trader C decides on bid based on intrinsic value.

5.8 Using Leaderboard

1. Trader A opens leaderboard:
   - Trader B: Portfolio $500
   - Trader C: Portfolio $450
   - Trader A: Portfolio $470
2. Trader A identifies which traders/pets to target for maximizing portfolio.

Clarifying Questions & Answers

1. Trader Behavior & Controls

Q1: Who controls the Traders?
A: Each Trader is controlled by a human participant (human acts as Trader). The participant can act as all three Traders during the demo.

Q2: Do Traders act simultaneously or turn-by-turn?
A: Traders operate simultaneously, but since one participant is controlling all of them, actions happen one at a time in any order chosen by the participant.

Q3: Does each Trader have a separate interface?
A: Yes. Each Trader has its own window/panel, simulating how multiple people could trade at once.

Q4: Can Traders see each other’s inventory or cash?
A: No. Each Trader sees only their own cash, locked cash (for active bids), and pets.

---

2. Market Mechanics & Trading Rules

Q5: Can Traders buy multiple pets at once?
A: Yes, both from the new supply and the market, as long as they have enough cash.

Q6: Can multiple bids exist for the same pet?
A: Only the highest bid counts. If a new higher bid comes in, it replaces the previous one and releases that bidder's locked cash.

Q7: Can a Trader bid on their own pets?
A: No — a Trader cannot buy their own pets.

Q8: Can bids be higher or lower than the asking price?
A: Yes, any bid is allowed. The seller chooses whether to accept or reject the bid.

Q9: What happens if a bid is withdrawn?
A: Locked cash is immediately released, and the seller is notified.

Q10: Can pets be relisted for sale?
A: Yes, but only after withdrawing the previous listing. Any active bids are automatically rejected when withdrawn.

Q11: Can a pet be listed multiple times simultaneously?
A: No. Each individual pet can have only one active listing, but a Trader can list multiple different pets at once.

Q12: Are trades instantaneous?
A: Yes. Once a seller accepts a bid, the transaction executes immediately and updates inventory and cash.

Q13: Are there any cash or bid restrictions?
A: Bids cannot exceed a Trader's available (unlocked) cash. Cash is locked for active bids and released when a bid is withdrawn or rejected.

---

3. Pets & Valuation

Q14: Are pets unique or generic?
A: Each pet is unique — its own age, health, and intrinsic value tracked individually.

Q15: How is age handled?
A: Age starts at 0 when purchased new and increases continuously, even if the pet is being traded.

Q16: How is health handled?

A: Health fluctuates ±5% at each valuation update.

Q17: How is intrinsic value calculated?

A: Same formula for all pets:

$$\text{Intrinsic Value} = \text{Base Price} \times \left(\frac{\text{Health}}{100}\right) \times \left(\frac{\text{Desirability}}{10}\right) \times \left(1 - \frac{\text{Age}}{\text{Lifespan}}\right)$$

Q18: Can expired pets be sold or bought?

A: Yes. Pets remain in inventory at zero intrinsic value, but traders can still bid on them.

Q19: What are the starting Base Prices for pets?

A: Each breed has a fixed retail price (e.g., Labrador = $100, Beagle = $90, Goldfish = $5, etc.)

Q20: Is the pet dictionary editable by participants?

A: No. The dictionary is read-only.

---

4. UI / Views

Q21: What should the market view show?

A: Current listings, asking price, most recent trade price, and new supply count. Default order = newest listings first.

Q22: What should the analysis view show?

A: Full fundamentals for each pet — age, health, desirability, maintenance, intrinsic value — for Traders to make decisions.

Q23: What does the leaderboard show?

A: Total portfolio value (cash + locked cash + market value). Updates in real-time as trades and valuations change.

Q24: What notifications are required?

A: Bid received, accepted, rejected, withdrawn, outbid. Each includes pet, price, and counterparty.

---

5. System Behavior

Q25: How often are valuations updated?

A: Every minute (or configurable). Variance ±5% on health/desirability.

Q26: What happens on UI refresh?

A: If any valuation has changed or a trade occurs, all relevant panels **update immediately.**

Q27: Can pets have multiple active bids?

A: No. Only one active bid per pet.

Q28: Are there limits on inventory or listings?

A: No. Traders can hold unlimited pets as long as they have cash, and list as many pets as they own.

Q29: How are transactions handled for multiple bids or relisted pets?

A: Highest bid replaces previous bids. Relisting requires withdrawing first, which cancels active bids.

UX / Experience Considerations left to Participant Decisions

Notes for Participants:

- Scoring may consider thoughtful design choices.
- The goal is to see participants **make intentional UX decisions** rather than follow a rigid template.

1. Market View Display

- How pets are **visually presented** (layout, card vs. table, icons, colors).
- Optional **sorting/filtering** (e.g., by type, price, age, health).
- Whether to highlight new listings, recently updated valuations, or expired pets.

2. Trader Panel / Inventory

- How inventory is **organized and displayed** (grouping, tabs, list vs. grid).
- How available cash, locked cash, and total portfolio value are shown.
- Whether to include **visual indicators for active bids**.

3. Analysis / Drill-Down View

- How the detailed pet fundamentals are presented (charts, tables, visual cues).
- Optional **highlighting of key metrics** (e.g., intrinsic value trends, age vs. lifespan).
- How easily a Trader can interpret information to decide on a bid.

4. Leaderboard

- How total portfolio values are shown (aggregate, per component, color-coded).

- Optional inclusion of visual cues for relative performance vs. other Traders.
- Placement and prominence of the leaderboard in the Trader's panel.

5. Notifications

- How alerts are displayed (pop-ups, banners, inline messages).
- Duration, visibility, or dismissal mechanics for notifications.
- Whether multiple notifications are grouped or stacked.

6. Bid / Trade Interaction

- How bidding is initiated and confirmed (modal, inline, drag-and-drop).
- How bids are visually distinguished (active, rejected, outbid).
- Optional UX cues for "locked cash" or unavailable funds.

7. Listing / Delisting Pets

- How sellers list pets for sale (form, drag-and-drop, quick-action buttons).
- How withdrawn or relisted pets are visually indicated.
- Optional prompts or confirmations for critical actions (accept/reject, withdrawal).

8. Valuation Updates

- How real-time updates (age, health, intrinsic value) are reflected in the UI.
- Optional visual cues for metric changes (animation, color changes, arrows).

9. Multiple Active Panels

- Whether participants **show all Trader panels at once** or allow toggling between Traders.
- Optional decision on **responsive layout for multiple simultaneous views**.

10. Optional UX Enhancements

- Any additional usability improvements that enhance clarity or reduce cognitive load (e.g., tooltips, hover info, color-coded risk indicators, highlighting expired pets).